

Games

- In AI, games serve as a fundamental area of research for understanding and testing decision-making capabilities. Games provide well-defined environments where AI systems can learn strategies, make decisions, and handle competition against other agents or human players.
- The study of optimal decision-making in games focuses on finding the best possible actions for an AI agent to maximize its chances of winning, often under the constraint that its opponent is also trying to do the same.

Game Theory and Decision Making in Games:

- Game theory is a branch of mathematics and economics that studies the behavior of decision-makers (or players) in strategic situations where the outcome depends on the choices of multiple factors.
- In the context of AI, game theory provides a foundation for developing algorithms that make optimal decisions by reasoning about possible actions and their consequences.

Type of Games:

- Zero-sum games: In these games, one player's gain is equivalent to another player's loss (e.g., chess, tic-tac-toe). The goal of AI here is to maximize its own reward while minimizing the opponent's potential reward.
- Non-zero-sum games: These games allow for cooperation, where players can both benefit from certain strategies (e.g., in trading or negotiations).

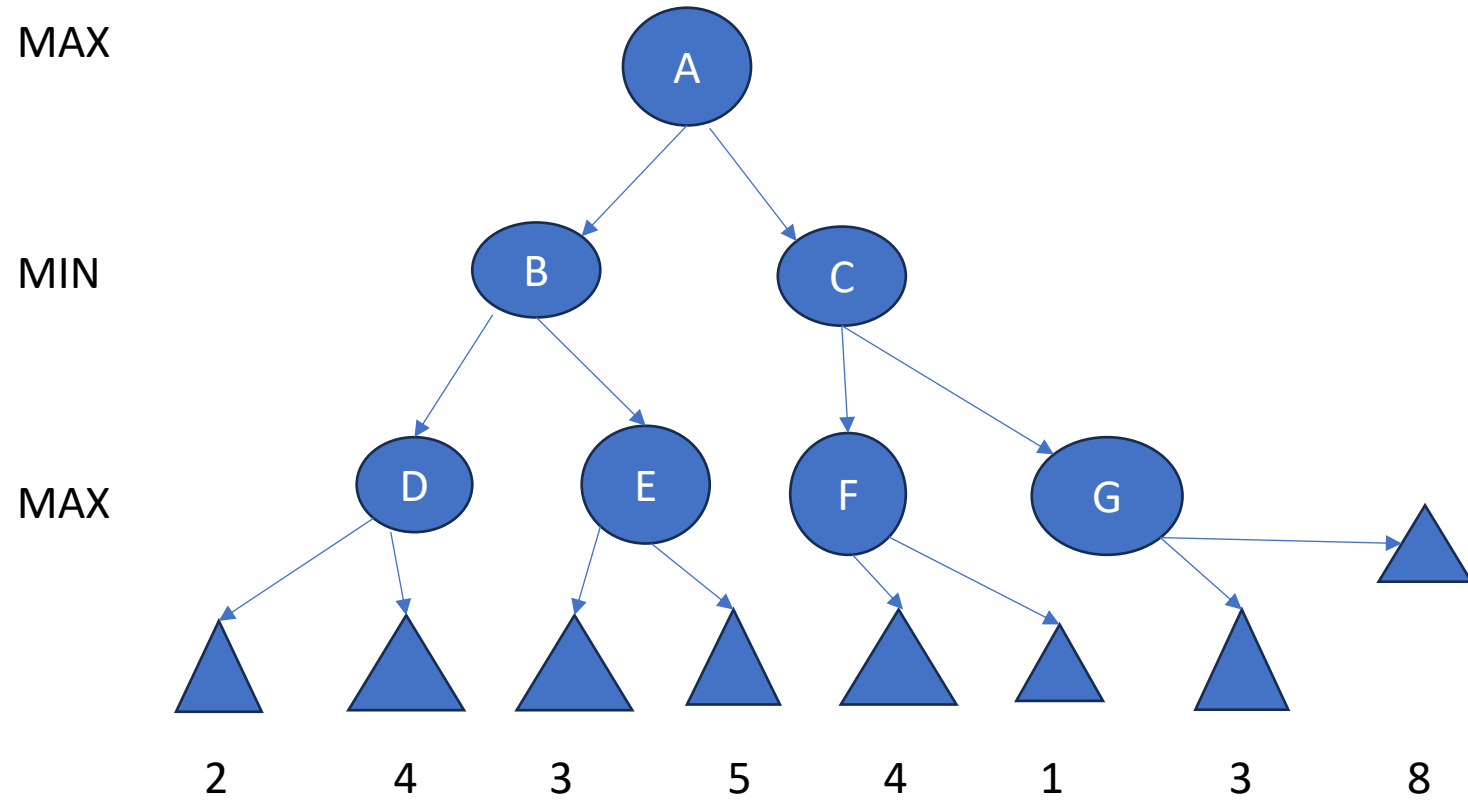
Optimal decision making in Games

- Optimal decision-making in games refers to choosing the best action at each stage of the game that leads to the most favorable outcome for the AI.
 - In turn-based games, this means maximizing the agent's winning chances while minimizing the opponent's winning chances.
 - AI systems achieve this by analyzing potential future game states and outcomes based on the current state of the game.
1. Game Trees: Many games can be modeled as a tree structure where each node represents a game state and each edge represents a possible move.
 - The AI evaluates the tree by simulating all possible future game states resulting from its and the opponent's actions.
 - The AI then picks the move leading to the best possible outcome based on these simulations.
 2. Evaluation Functions: To evaluate game states, AI systems use evaluation functions. These functions assign a numeric value to each state, representing how favorable it is for the player.
 - For example, in chess, an evaluation function might give a score based on the number and type of pieces each player has, the control of the board, and best move.

3. Search Algorithms: AI uses search algorithms to explore the game tree and find optimal moves.
- Minimax Algorithm: The most common approach for making optimal decisions in two-player zero-sum games is the Minimax algorithm. It assumes that both players are playing optimally.
 - Maximizing player (AI): Chooses the move that leads to the best possible outcome for itself.
 - Minimizing player (opponent): Chooses the move that leads to the worst outcome for the AI, minimizing the AI's advantage.

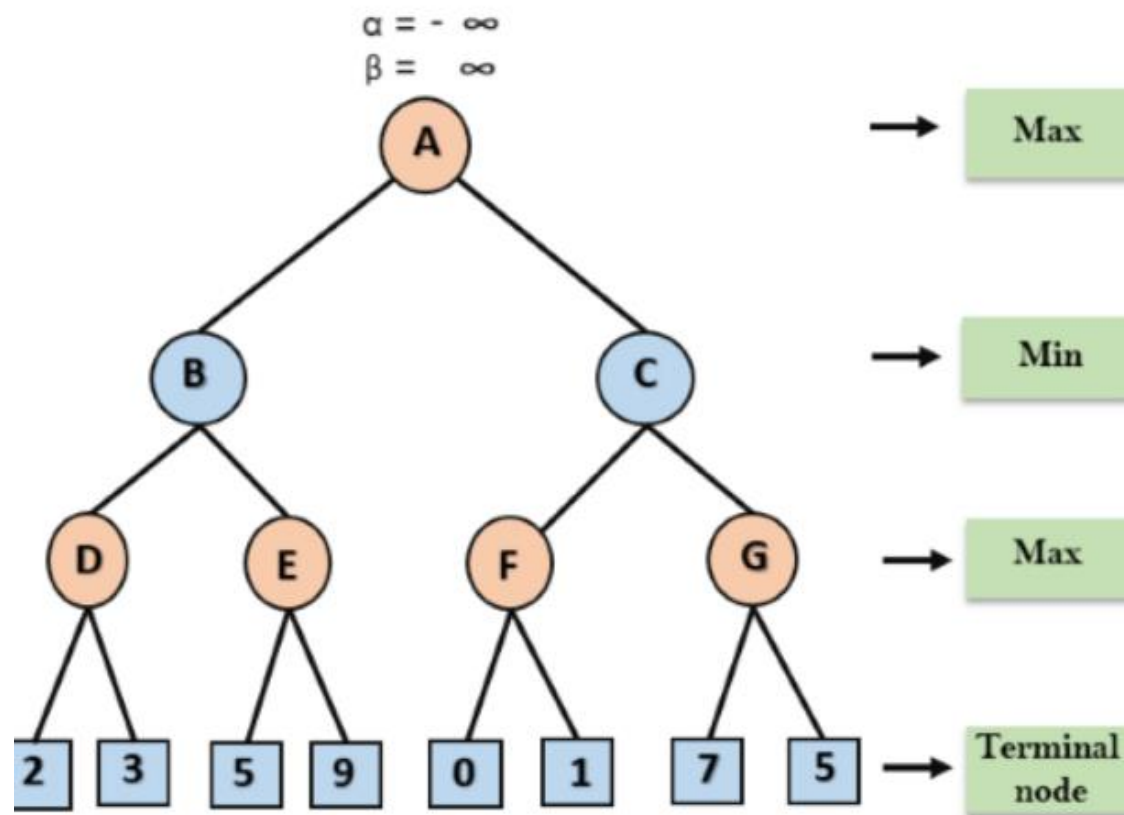
MiniMax Algorithm

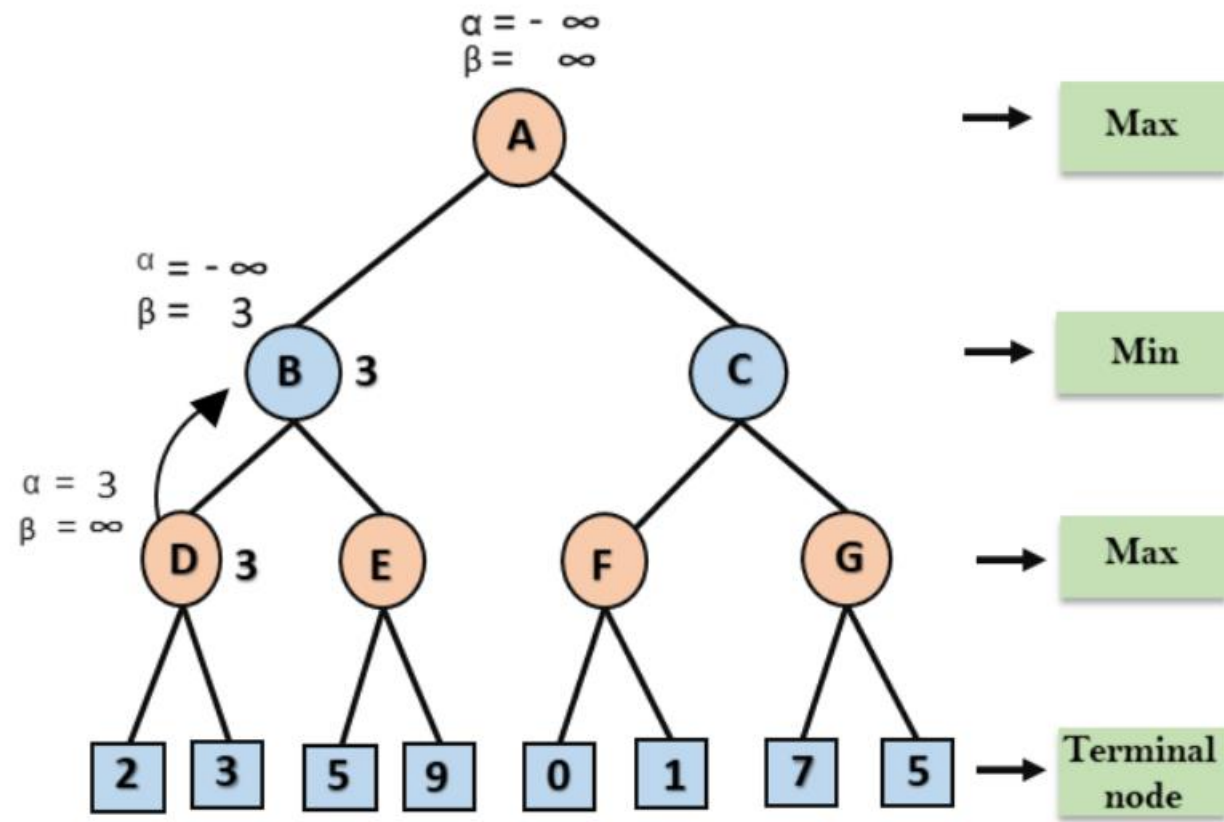
- It is a backtracking algorithm.
- It use best move strategy.
- Max will try to maximize its utility / reward(best move).
- Min will try to minimize utility(worst move).
- Time complexity: $O(b^d)$ where, b = branching factor and d =depth tree.
- Example: In chess 35 choices is possible and no of move that is depth is 50 and for two player the depth will be 100 so $(35)^{100}$ no of nodes. Therefore in this type of case we will use alpha - beta pruning.

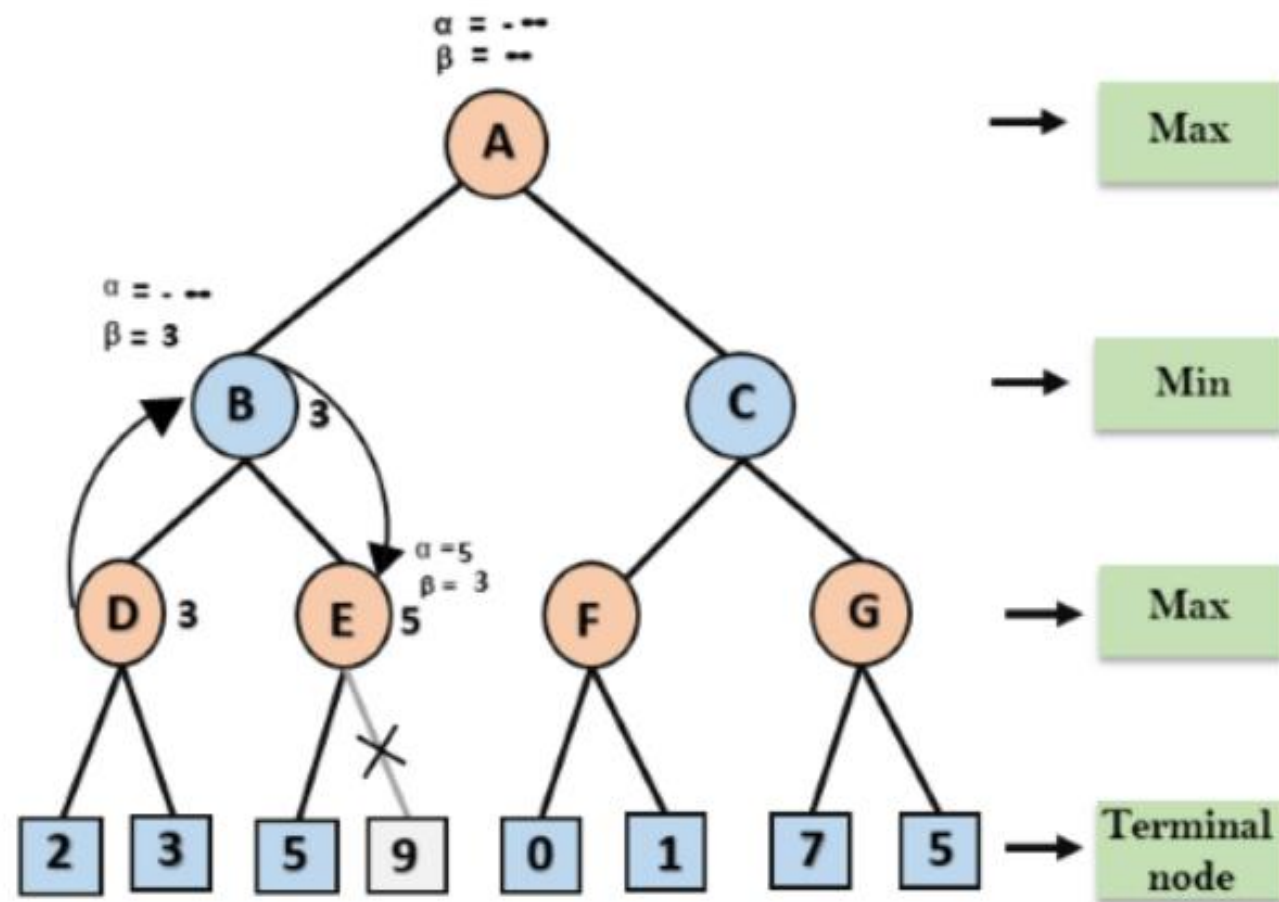


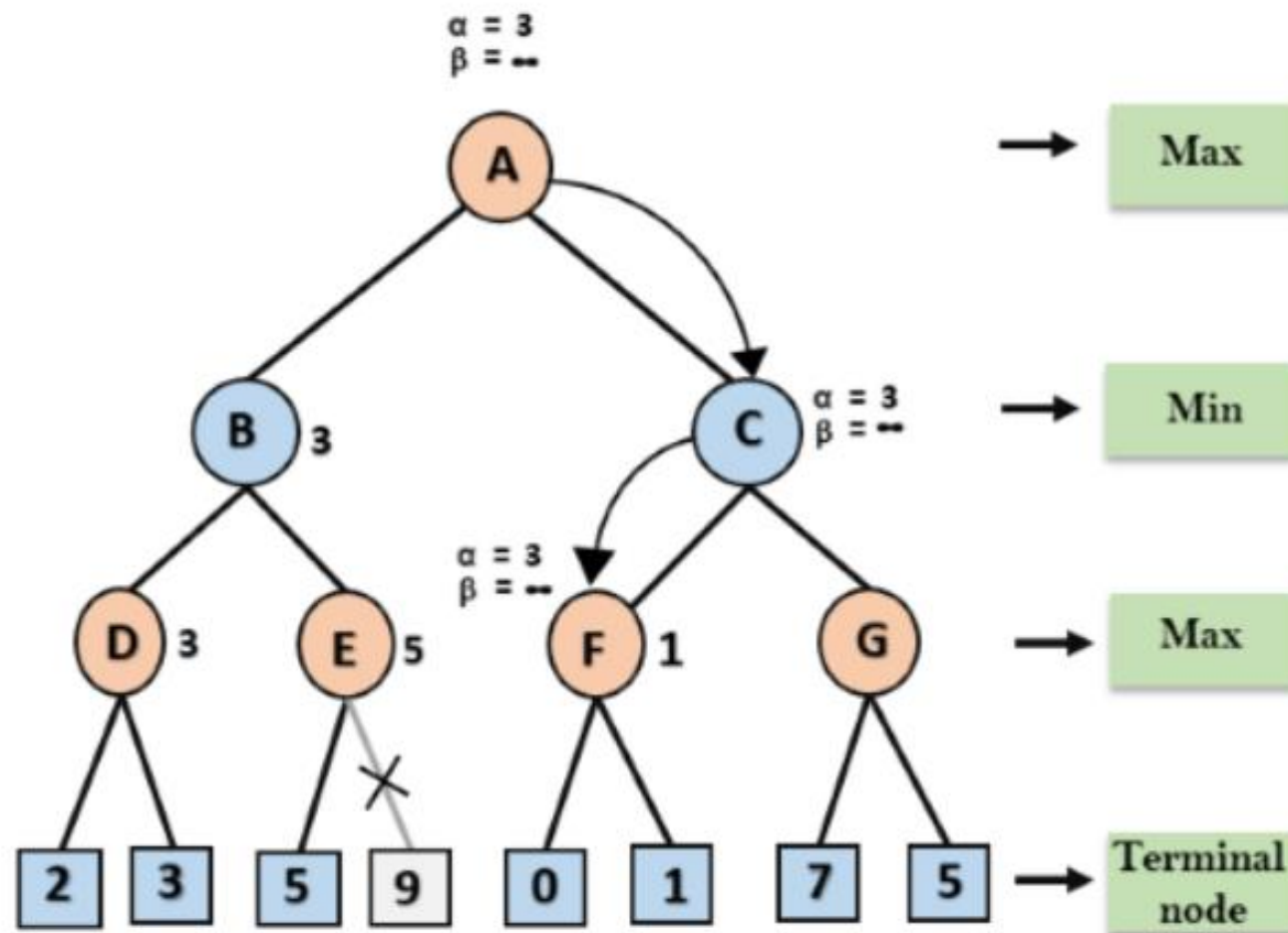
Alpha – Beta pruning

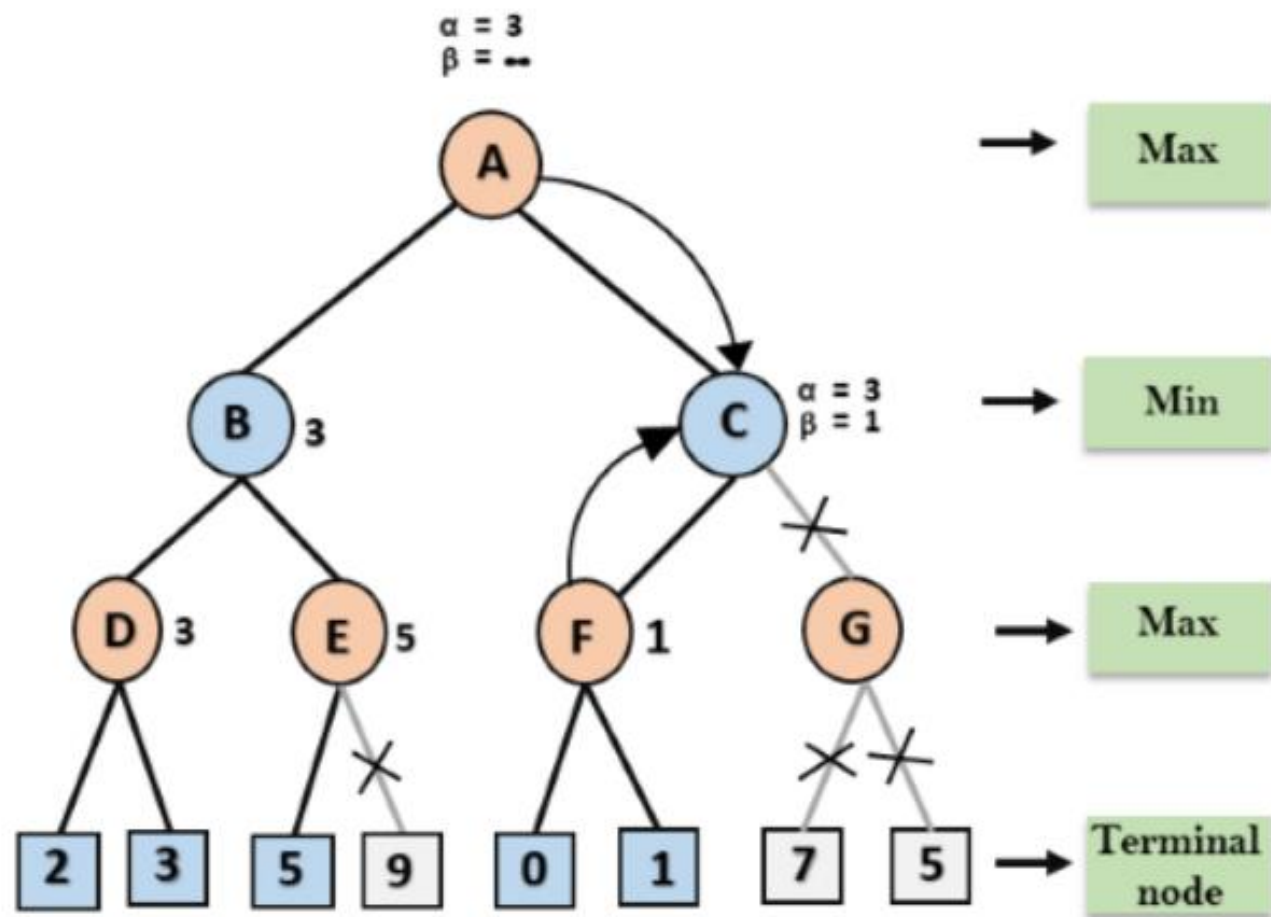
- Cut – off search by exploring a less number of nodes.
- As we have seen in the minimax search algorithm that the number of game states it has to examine are exponential in depth of the tree. Since we cannot eliminate the exponent, but we can cut it to half. Hence there is a technique by which without checking each node of the game tree we can compute the correct minimax decision, and this technique is called pruning. This involves two threshold parameter Alpha and beta for future expansion, so it is called alpha-beta pruning. It is also called as Alpha-Beta Algorithm.
- Alpha: The best (highest-value) choice we have found so far at any point along the path of Maximizer. The initial value of alpha is $-\infty$.
- Beta: The best (lowest-value) choice we have found so far at any point along the path of Minimizer. The initial value of beta is $+\infty$.
- The main condition which required for alpha-beta pruning is: $\alpha \geq \beta$.
- The Max player will only update the value of alpha.
- The Min player will only update the value of beta.
- While backtracking the tree, the node values will be passed to upper nodes instead of values of alpha and beta.
- We will only pass the alpha, beta values to the child nodes.

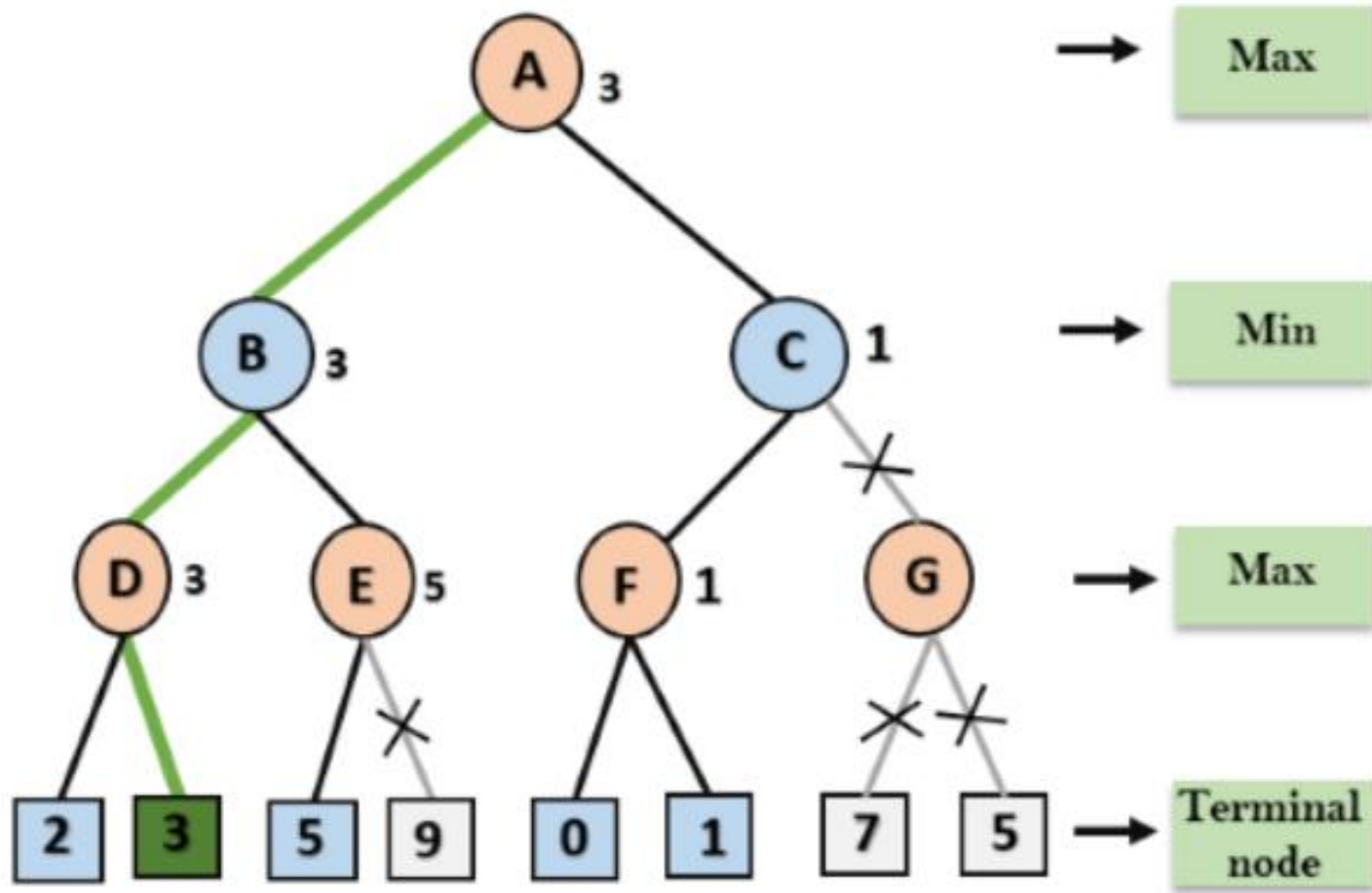






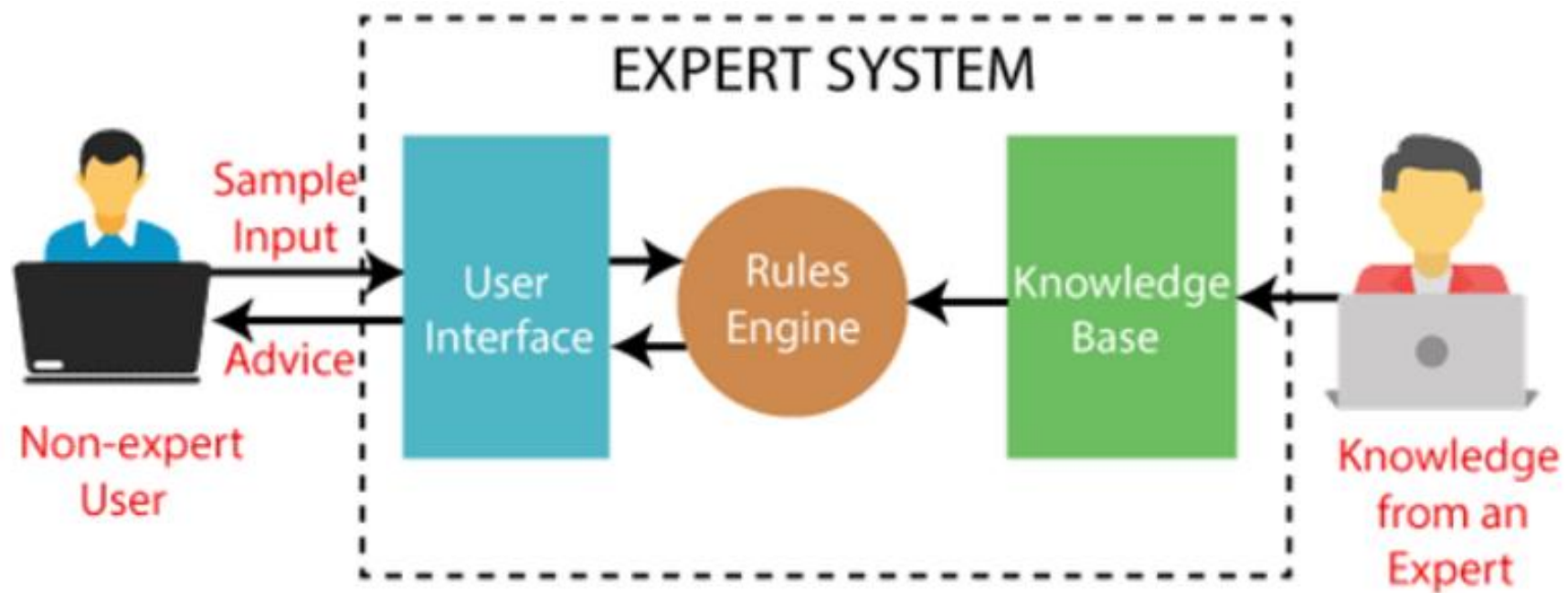






Expert system

- An expert system is a computer program that is designed to solve complex problems and to provide decision-making ability like a human expert.
- It performs this by extracting knowledge from its knowledge base using the reasoning and inference rules according to the user queries.
- The expert system is a part of AI, and the first ES was developed in the year 1970, which was the first successful approach of artificial intelligence.
- It solves the most complex issue as an expert by extracting the knowledge stored in its knowledge base. The system helps in decision making for complex problems using **both facts and heuristics like a human expert**.
- It is called so because it contains the expert knowledge of a specific domain and can solve any complex problem of that particular domain. These systems are designed for a specific domain, such as medicine, science, etc.
- The performance of an expert system is based on the expert's knowledge stored in its knowledge base. The more knowledge stored in the KB, the more that system improves its performance.
- One of the common examples of an ES is a suggestion of spelling errors while typing in the Google search box.

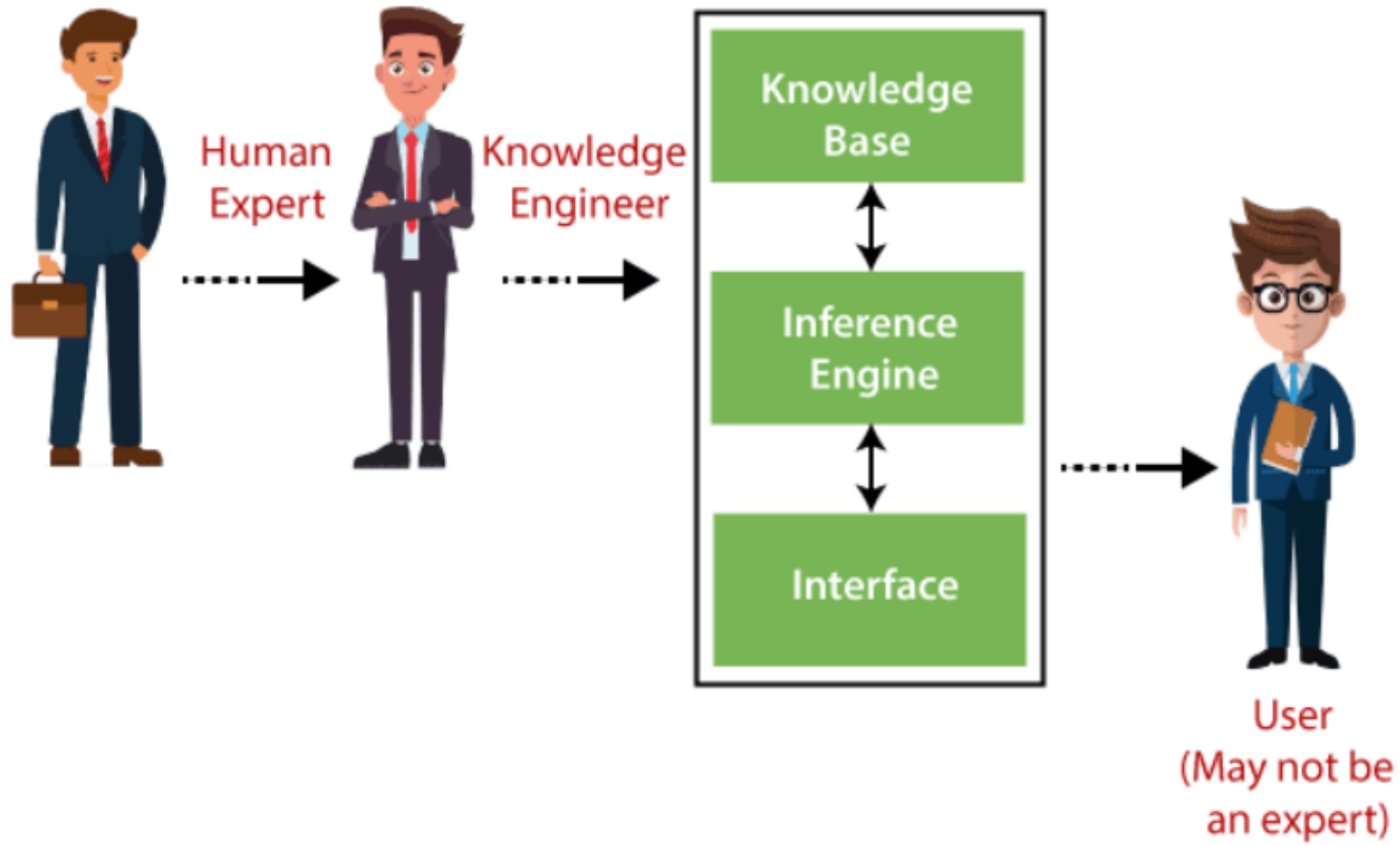


Characteristics of Expert System

- **High Performance:** The expert system provides high performance for solving any type of complex problem of a specific domain with high efficiency and accuracy.
- **Understandable:** It responds in a way that can be easily understandable by the user. It can take input in human language and provides the output in the same way.
- **Reliable:** It is much reliable for generating an efficient and accurate output.
- **Highly responsive:** ES provides the result for any complex query within a very short period of time.

Components of Expert System

- An expert system mainly consists of three components:
 1. User Interface
 2. Inference Engine
 3. Knowledge Base



1. User Interface

With the help of a user interface, the expert system interacts with the user, takes queries as an input in a readable format, and passes it to the inference engine. After getting the response from the inference engine, it displays the output to the user. In other words, it is an interface that helps a non-expert user to communicate with the expert system to find a solution.

2. Inference Engine(Rules of Engine)

The inference engine is known as the brain of the expert system as it is the main processing unit of the system. It applies inference rules to the knowledge base to derive a conclusion or deduce new information. It helps in deriving an error-free solution of queries asked by the user.

With the help of an inference engine, the system extracts the knowledge from the knowledge base.

There are two types of inference engine:

- **Deterministic Inference engine:** The conclusions drawn from this type of inference engine are assumed to be true. It is based on facts and rules.
- **Probabilistic Inference engine:** This type of inference engine contains uncertainty in conclusions, and based on the probability.

Inference engine uses the below modes to derive the solutions:

- **Forward Chaining:** It starts from the known facts and rules, and applies the inference rules to add their conclusion to the known facts.
- **Backward Chaining:** It is a backward reasoning method that starts from the goal and works backward to prove the known facts.

3. Knowledge Base

The knowledge base is a type of storage that stores knowledge acquired from the different experts of the particular domain. It is considered as big storage of knowledge. The more the knowledge base, the more precise will be the Expert System.

It is similar to a database that contains information and rules of a particular domain or subject.

One can also view the knowledge base as collections of objects and their attributes.

Example: Such as a Lion is an object and its attributes are it is a mammal, it is not a domestic animal, etc.

Components of Knowledge Base

- **Factual Knowledge:** The knowledge which is based on facts and accepted by knowledge engineers comes under factual knowledge.
- **Heuristic Knowledge:** This knowledge is based on practice, the ability to guess, evaluation, and experiences.
- **Knowledge Representation:** It is used to formalize the knowledge stored in the knowledge base using the If-else rules.
- **Knowledge Acquisitions:** It is the process of extracting, organizing, and structuring the domain knowledge, specifying the rules to acquire the knowledge from various experts, and store that knowledge into the knowledge base.

There are three primary participants in the building of Expert System:

- 1.Expert:** The success of an ES much depends on the knowledge provided by human experts. These experts are those persons who are specialized in that specific domain.
- 2.Knowledge Engineer:** Knowledge engineer is the person who gathers the knowledge from the domain experts and then codifies that knowledge to the system according to the formalism.
- 3.End-User:** This is a particular person or a group of people who may not be experts, and working on the expert system needs the solution or advice for his queries, which are complex.

- 1. No memory Limitations:** It can store as much data as required and can memorize it at the time of its application. But for human experts, there are some limitations to memorize all things at every time.
- 2. High Efficiency:** If the knowledge base is updated with the correct knowledge, then it provides a highly efficient output, which may not be possible for a human.
- 3. Expertise in a domain:** There are lots of human experts in each domain, and they all have different skills, different experiences, and different skills, so it is not easy to get a final output for the query. But if we put the knowledge gained from human experts into the expert system, then it provides an efficient output by mixing all the facts and knowledge
- 4. Not affected by emotions:** These systems are not affected by human emotions such as fatigue, anger, depression, anxiety, etc.. Hence the performance remains constant.
- 5. High security:** These systems provide high security to resolve any query.
- 6. Considers all the facts:** To respond to any query, it checks and considers all the available facts and provides the result accordingly. But it is possible that a human expert may not consider some facts.
- 7. Regular updates improve the performance:** If there is an issue in the result provided by the expert systems, we can improve the performance of the system by updating the knowledge base.

Advantages of Expert System

- These systems are highly reproducible.
- They can be used for risky places where the human presence is not safe.
- Error possibilities are less if the KB contains correct knowledge.
- The performance of these systems remains steady as it is not affected by emotions, tension, or fatigue.
- They provide a very high speed to respond to a particular query.

Limitations of Expert System

- The response of the expert system may get wrong if the knowledge base contains the wrong information.
- Like a human being, it cannot produce a creative output for different scenarios.
- Its maintenance and development costs are very high.
- Knowledge acquisition for designing is much difficult.
- For each domain, we require a specific ES, which is one of the big limitations.
- It cannot learn from itself and hence requires manual updates.

Applications of Expert System

- In designing and manufacturing domain: It can be broadly used for designing and manufacturing physical devices such as camera lenses and automobiles.
- In the knowledge domain: These systems are primarily used for publishing the relevant knowledge to the users. The two popular ES used for this domain is an advisor and a tax advisor.
- In the finance domain: In the finance industries, it is used to detect any type of possible fraud, suspicious activity, and advise bankers that if they should provide loans for business or not.
- In the diagnosis and troubleshooting of devices: In medical diagnosis, the ES system is used, and it was the first area where these systems were used.
- Planning and Scheduling: The expert systems can also be used for planning and scheduling some particular tasks for achieving the goal of that task.